

DTU



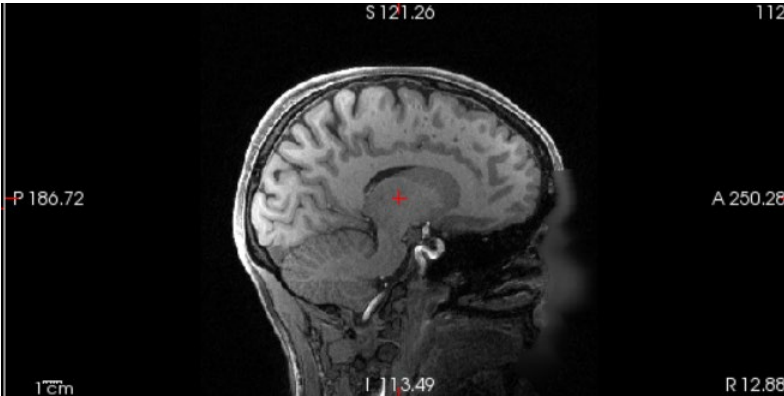
Course 22525

Surface-based metrics

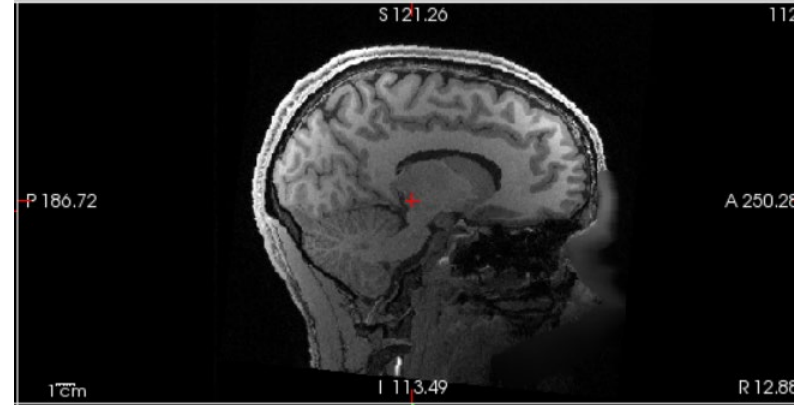
Axel Thielscher

Motivation: Limitation of volume-based approaches

Subject 1



Subject 2



Background: Gray matter (cortex) of the brain contains neuron cell bodies and dendrites (i.e., is the place where the computations are happening). Variations in gray matter thickness and cortical folding pattern indicate changes due to training, ageing and diseases → of interest in many neuroimaging studies.

Example task: Compare the gray matter (cortex) thickness of subject 1 and 2

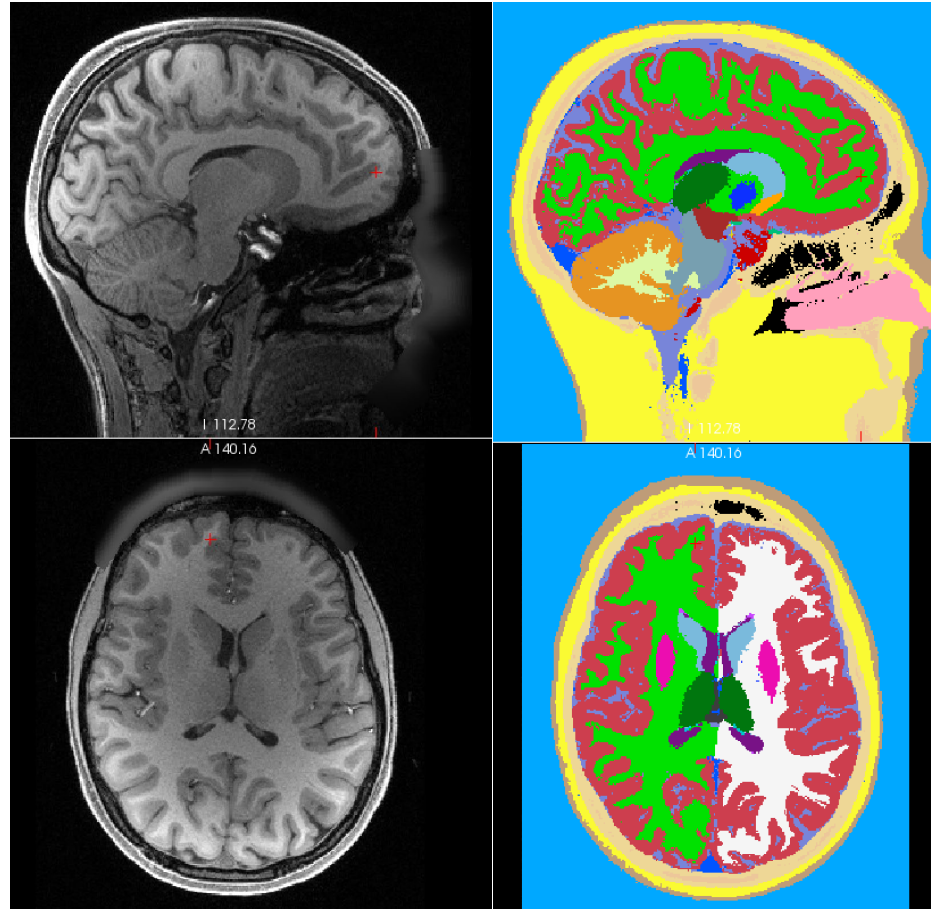
Approach:

- 1) Segment their gray matter
- 2) Measure thickness locally
- 3) Register to each other or to a common space to “overlay” the two images and compare

Motivation: Limitation of volume-based approaches

- 1) Segment their gray matter
... looks quite OK...

Subject 1



Puonti et al., Neuroimage, 2020

Motivation: Limitation of volume-based approaches

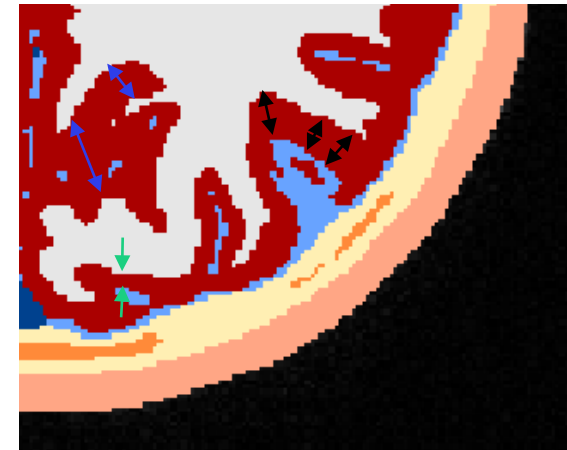
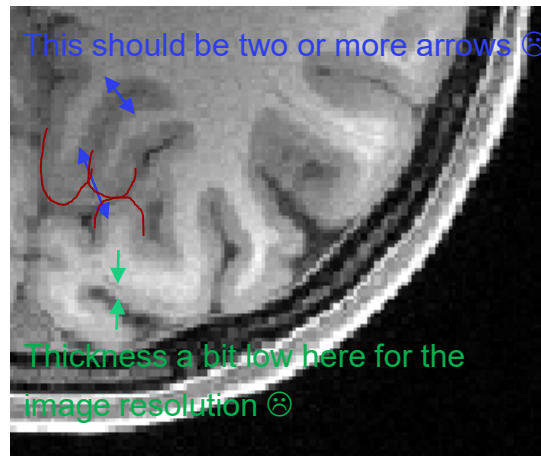
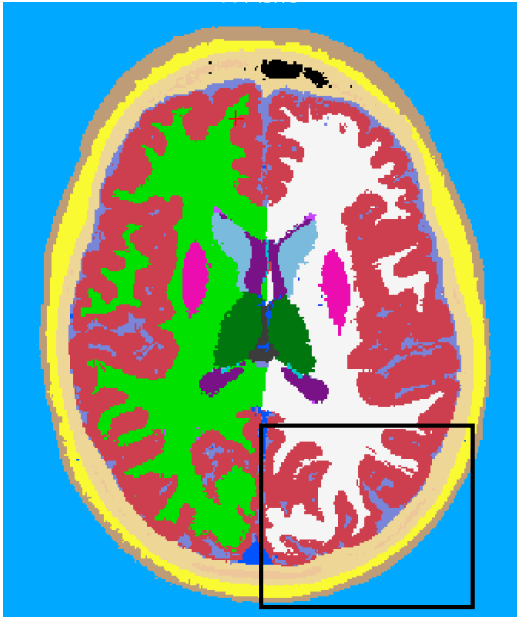
Subject 1

2) Make a method to measure thickness locally

... maybe still a bit noisy ...

... let's try some polishing (here: morphological operations)

... now get thickness at each gray matter position



Challenges:

- Partial volume effects
- Sensitivity to local noise

Motivation: Limitation of volume-based approaches

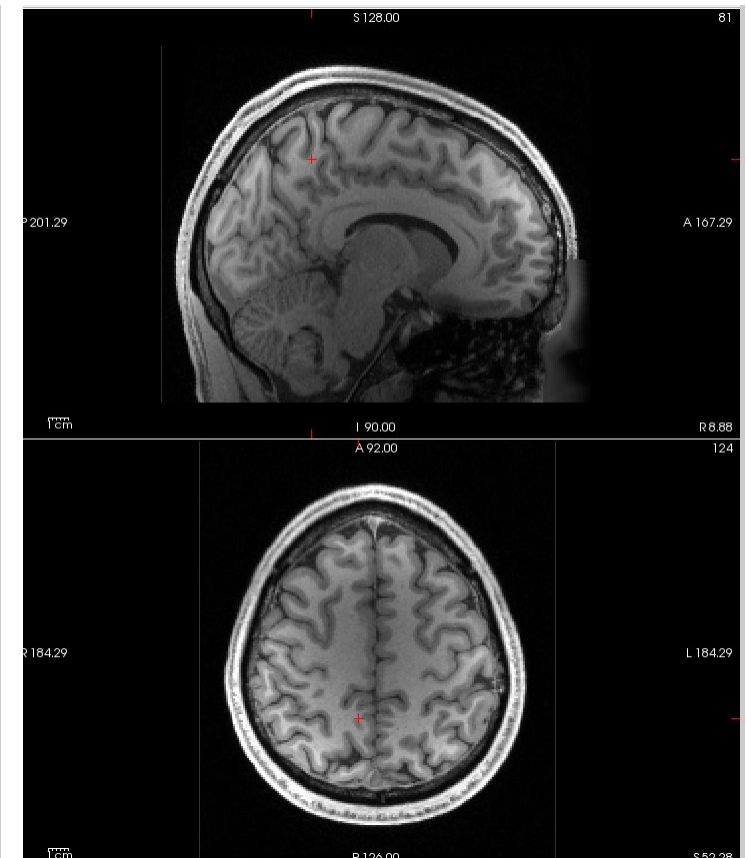
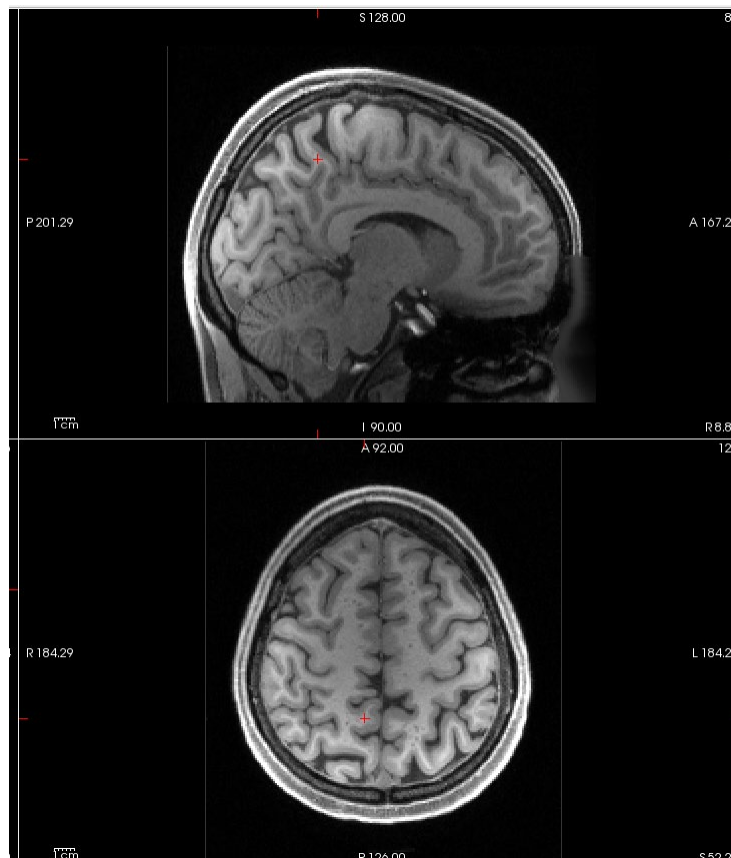
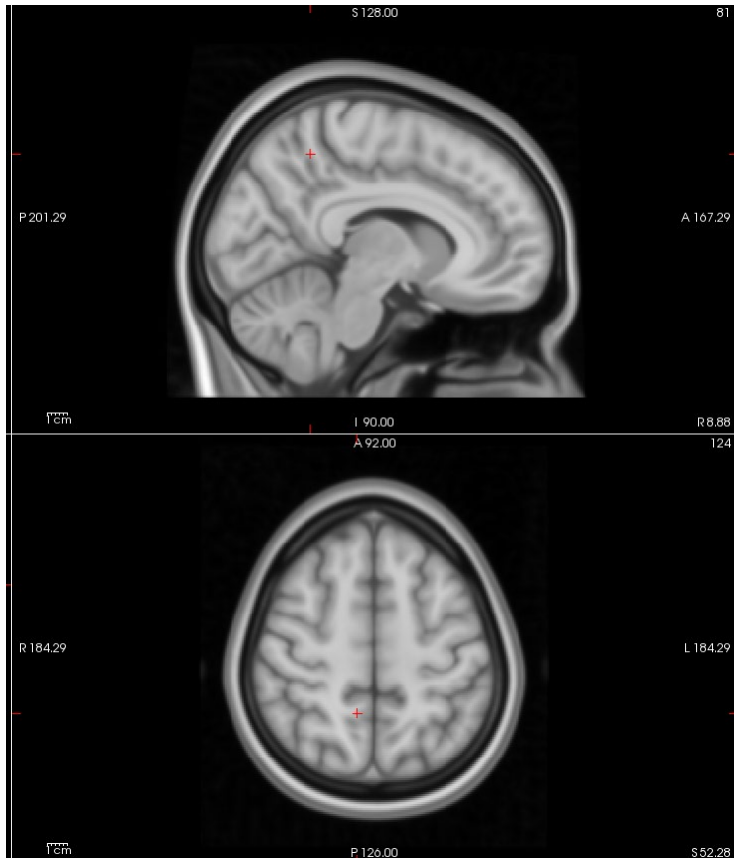
3) Register to each other or to a common space to “overlay” the two images to compare

Quite OK, right?

Group template (MNI)

Subject 1 in “MNI space”

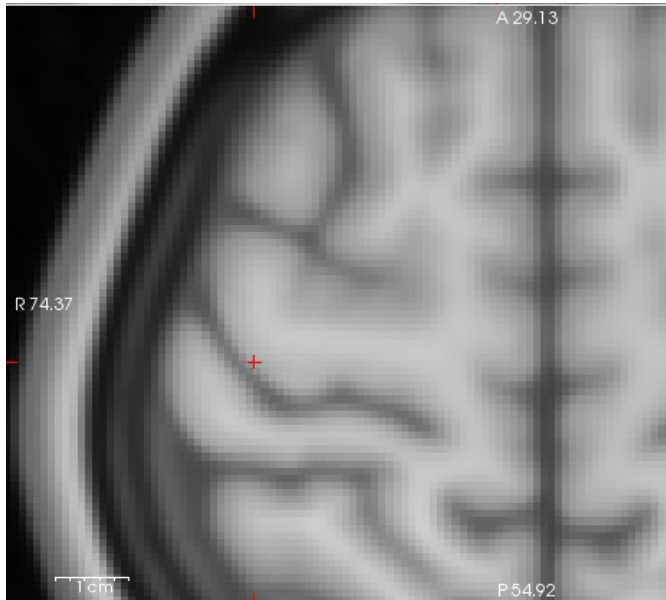
Subject 2 in “MNI space”



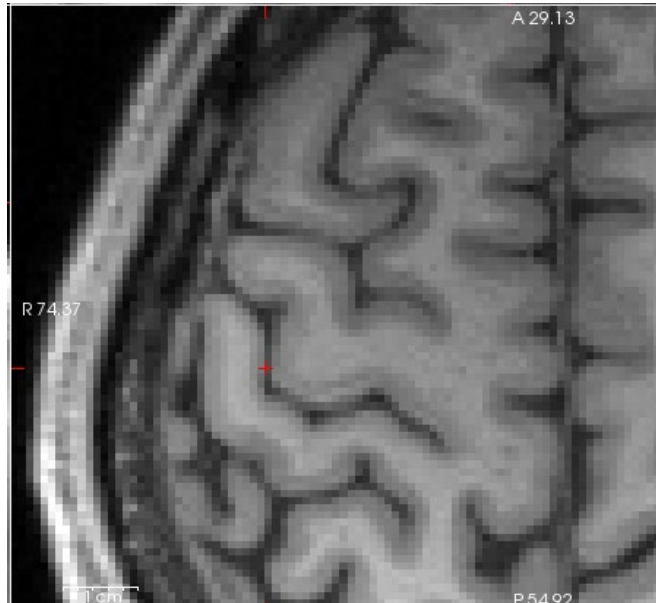
Motivation: Limitation of volume-based approaches

3) Register to each other or to a common space to “overlay” the two images to compare

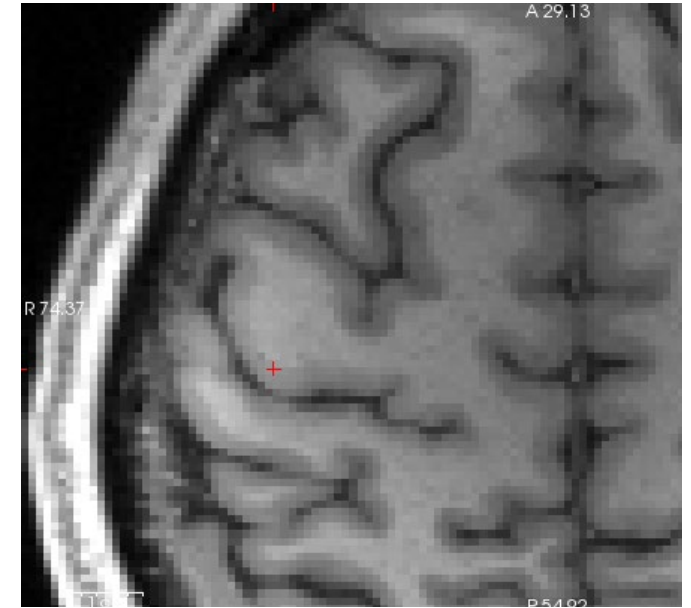
Group template (MNI)



Subject 1 in “MNI space”



Subject 2 in “MNI space”



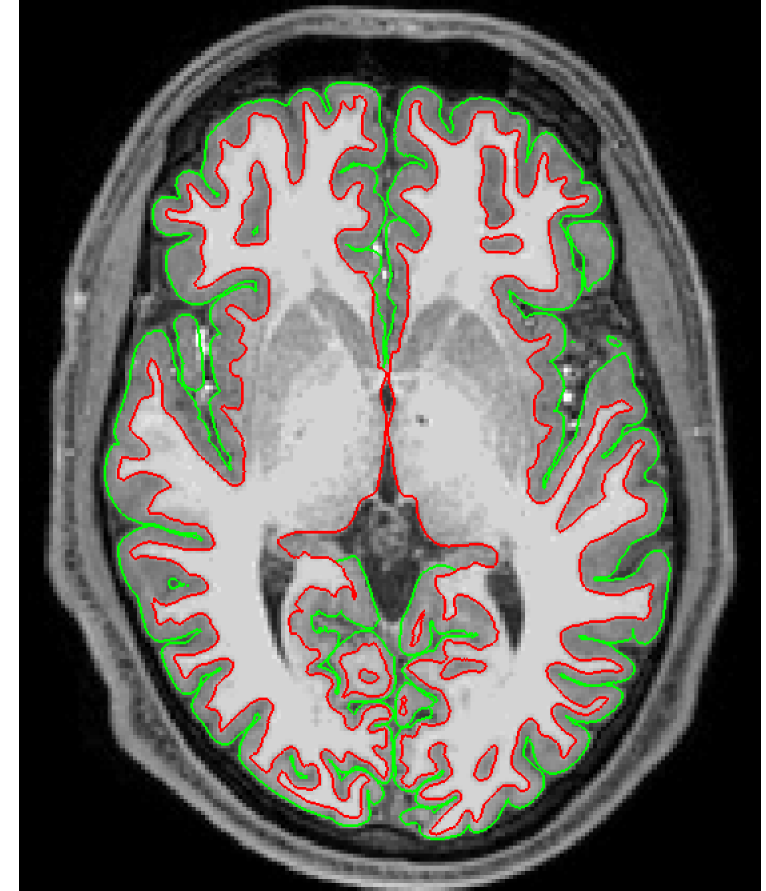
Quite good, but still mixes neighboring tissue types!

... but we need to accurately overlay gray matter onto each other for comparisons of local gray matter thickness ☹

Surface-based approaches

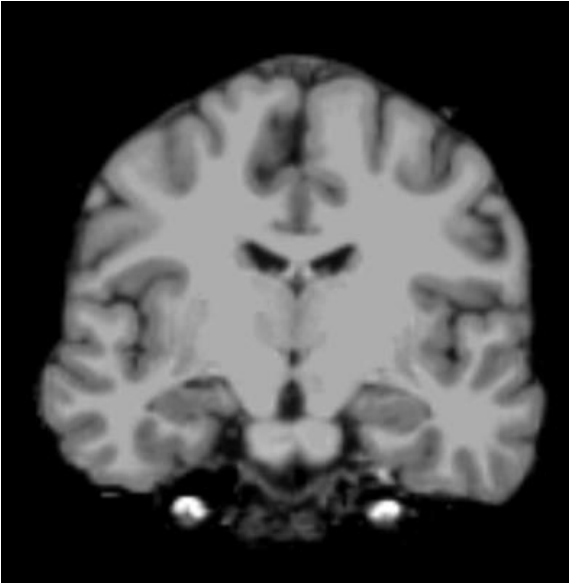
Aim:

- 1) Use surfaces to embed the knowledge into the thickness estimation that
 - the boundaries between gray matter and the neighboring tissues are smooth → ameliorate partial volume effects by interpolation
 - inner and outer boundaries always appear together → ensure topological correctness of the reconstructed gray matter sheet
- 2) Use surfaces to embed the knowledge into the registration that gray matter has to be registered on gray matter



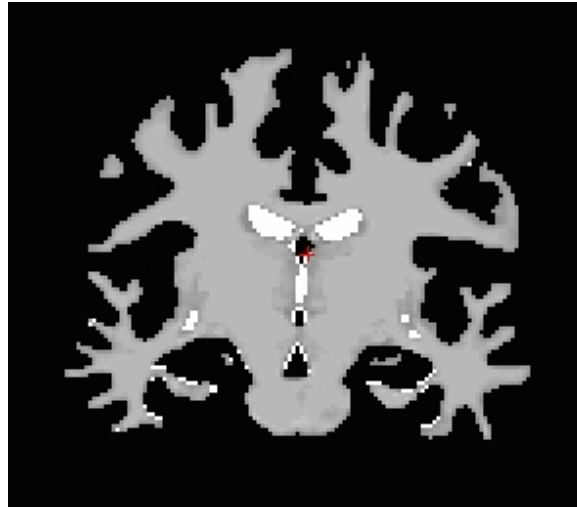
Surface Reconstruction of the brain

Bias corrected MRI,
brain mask applied

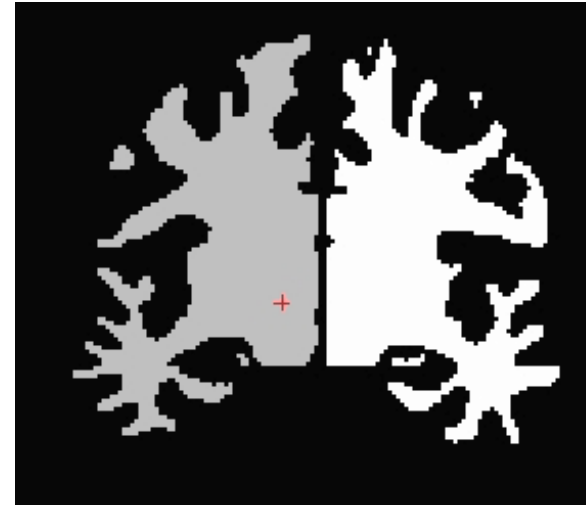


FreeSurfer tutorial

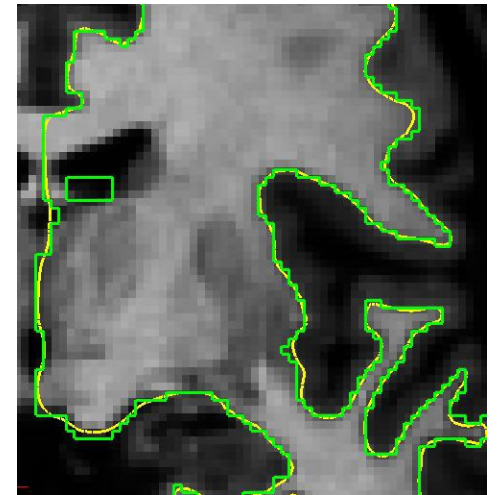
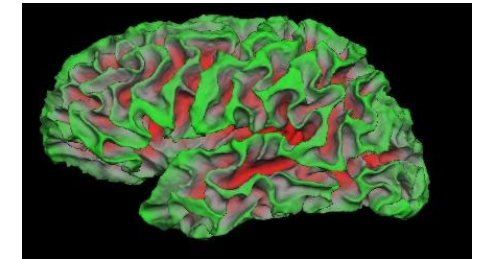
Get binary white matter mask,
fill inner "stuff"



Separate the hemispheres



White matter surface:
Reconstruct, fix topology
+ smooth



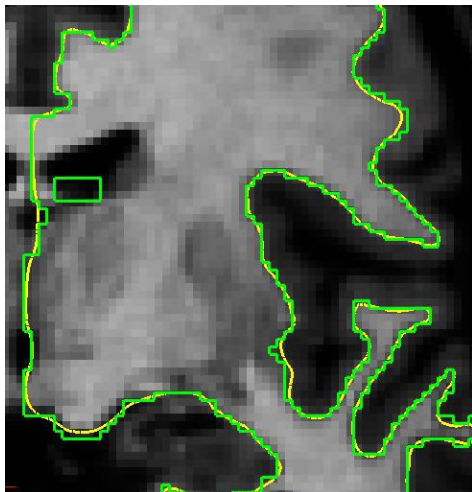
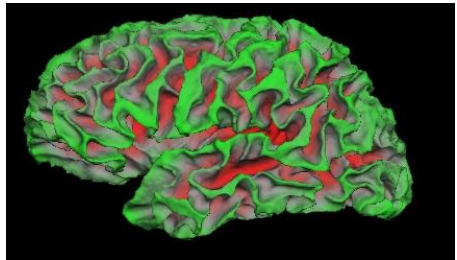
Note: Closed surface enables unambiguous definition of
inside/outside and of surface curvature

Red: concave parts

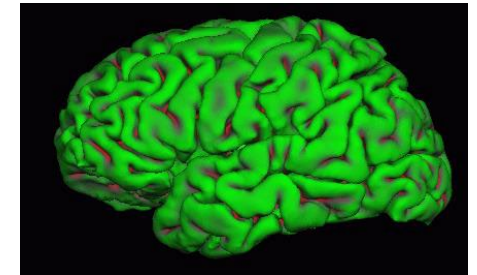
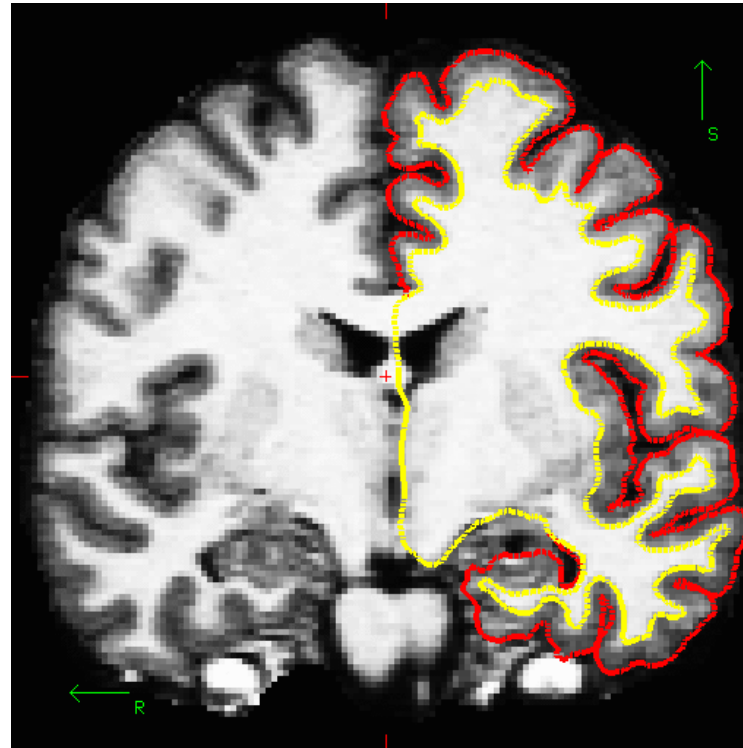
Green: convex parts

Surface Reconstruction of the brain

White matter surface



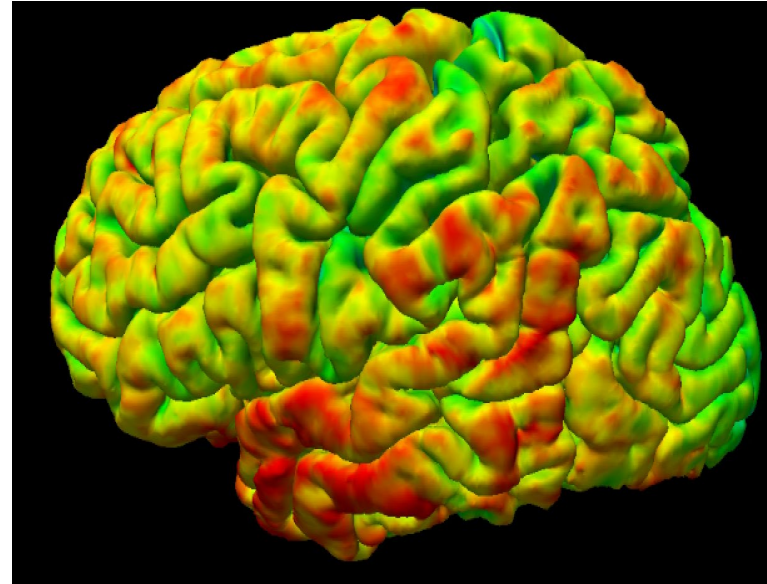
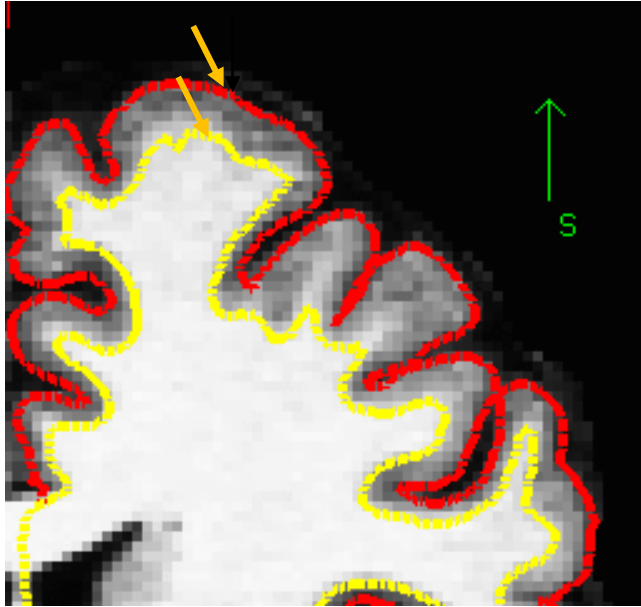
Grow surface until CSF is reached
(avoid self-intersections)



FreeSurfer tutorial

Measuring the distance between two surfaces

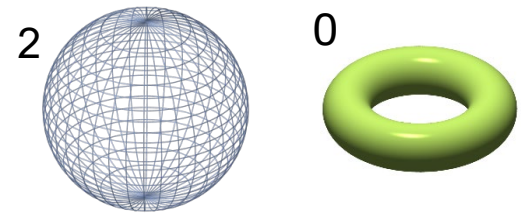
Same vertices on original (inner) and grown surface: Euclidean distance between vertex pairs is a good approximation of the cortical thickness



FreeSurfer tutorial

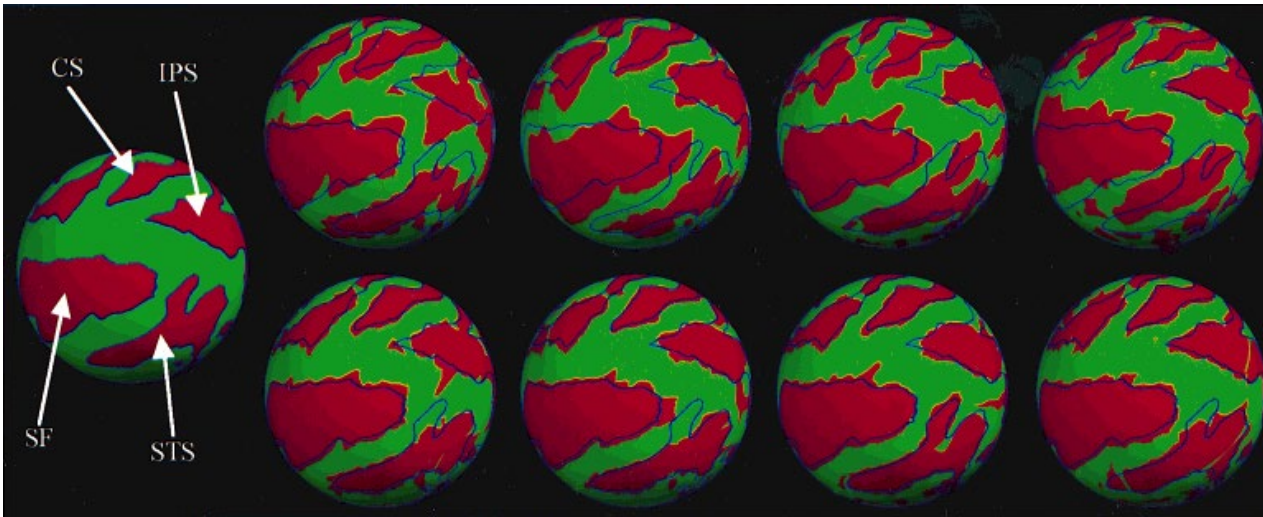
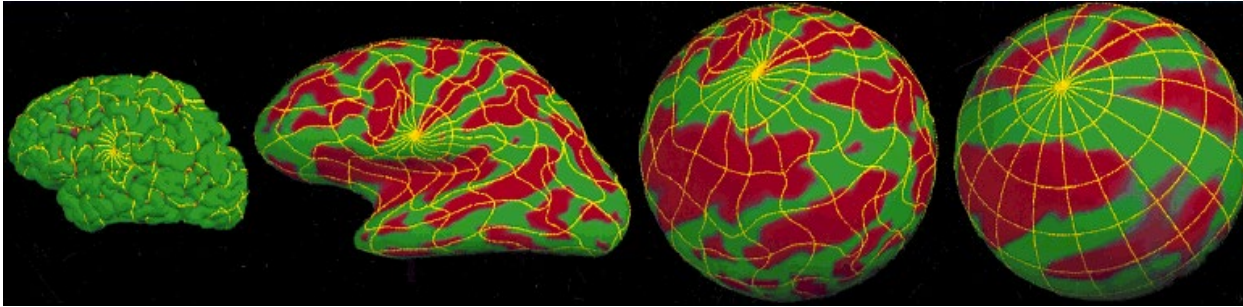
Spherical registration

Example ECs:



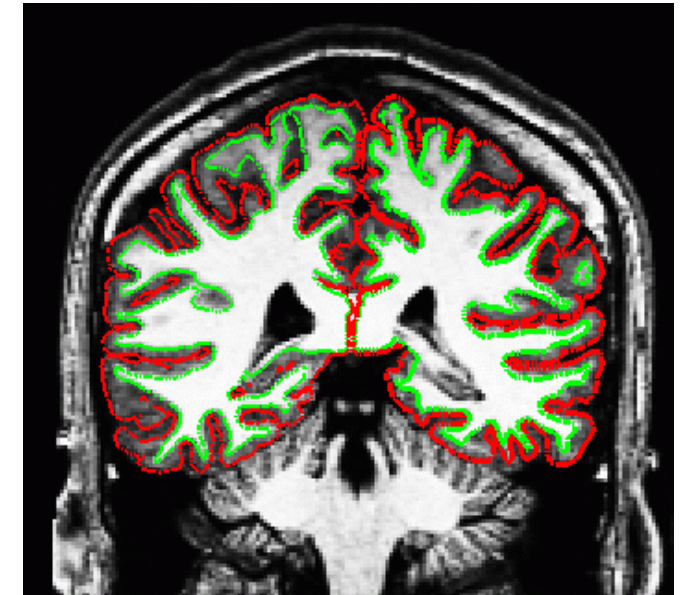
1) Expand cortical surface to a sphere
(Note: works only for surfaces with Euler characteristic 2)

2) Distort the sphere so that the pattern of positive and negative curvatures fits best to the pattern of another subject or the template cortical surface



Before
Registration

After
Registration



Fischl et al

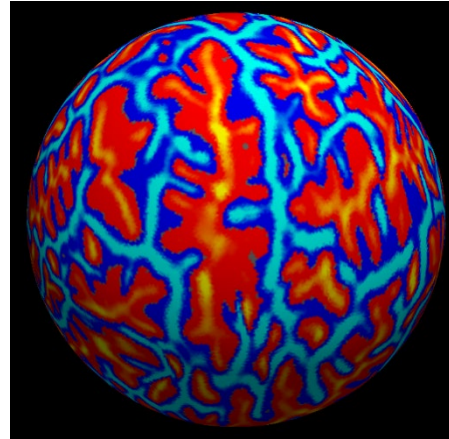
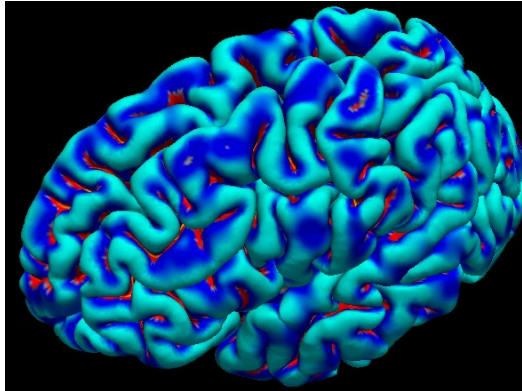
Human Brain Mapping, 1999, 8:272–284

Spherical registration

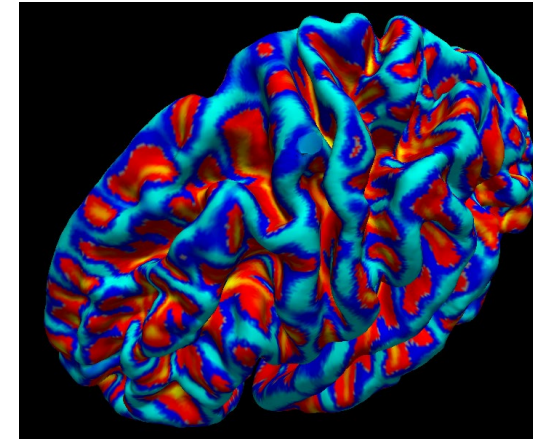
after curvature alignment

- 1) Expand cortical surface to a sphere
- 2) Distort the sphere so that the pattern of positive and negative curvatures fits best to the pattern of a template cortical surface

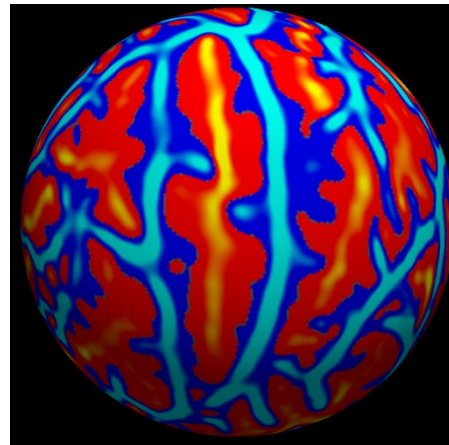
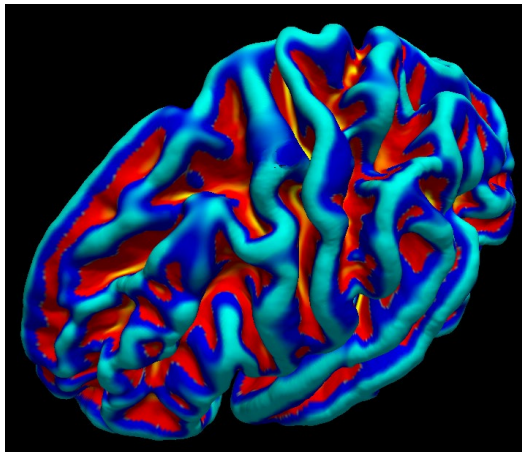
Subject



Subject curvature on template



Template
(fsaverage)

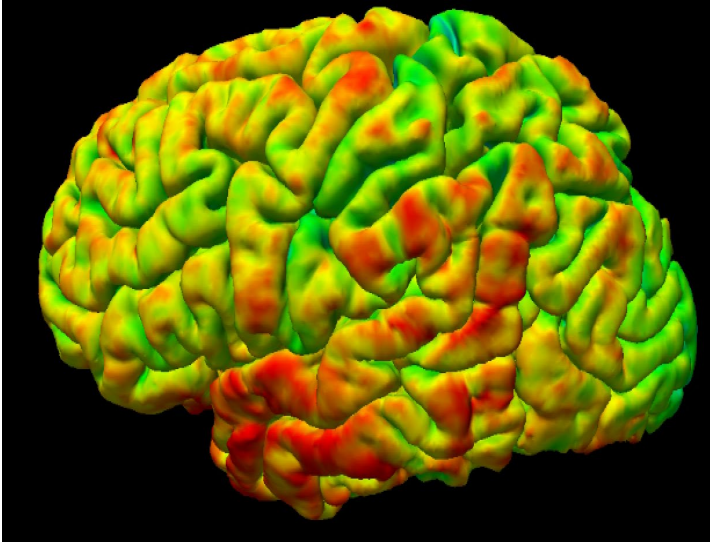


Fischl et al

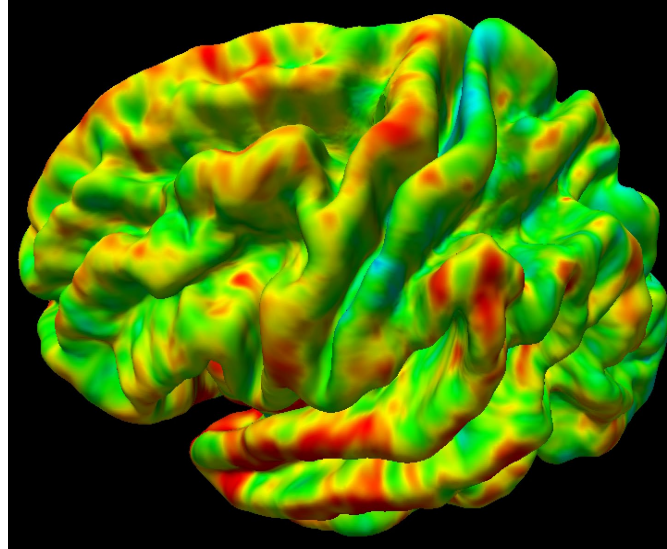
Human Brain Mapping, 1999, 8:272–284

Cortical thickness on template surface

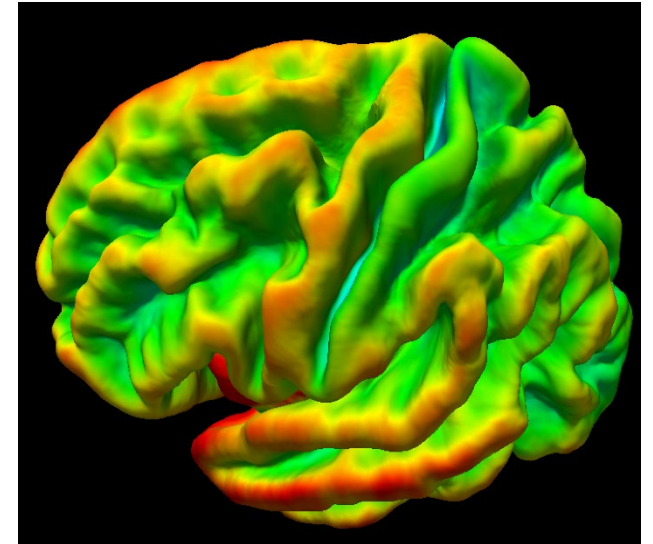
Subject 1: Cortical thickness on individual surface



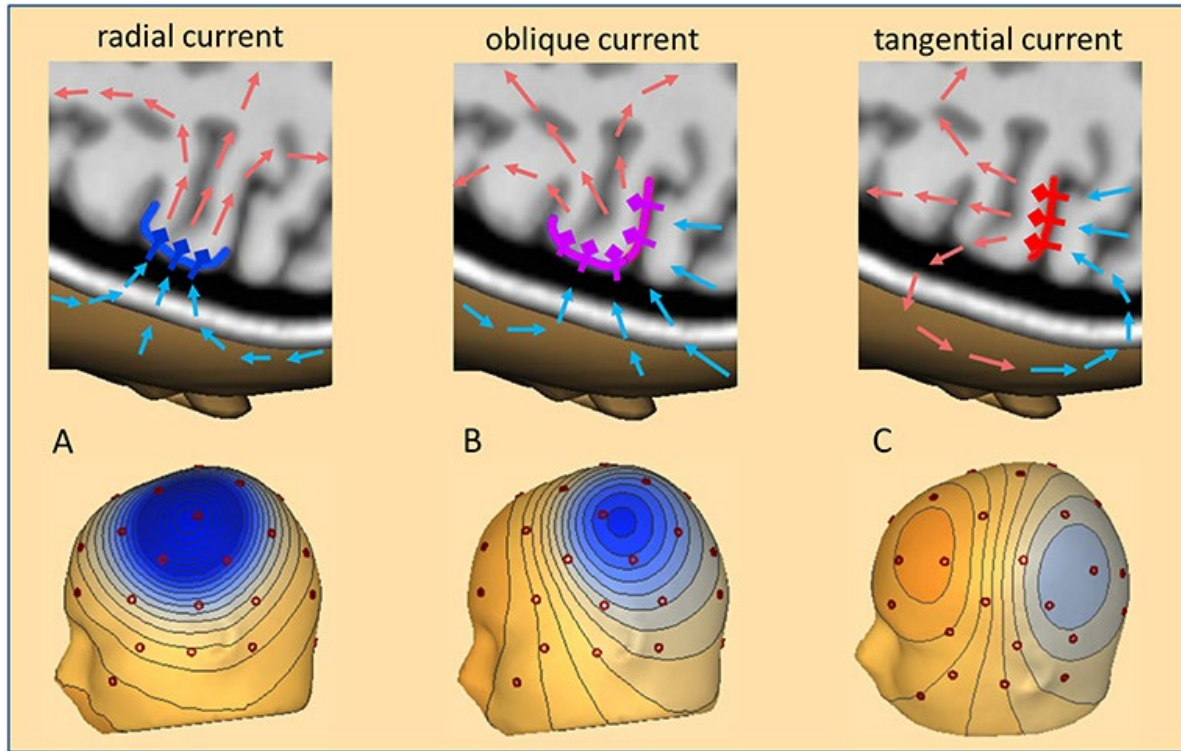
Subject 1: Cortical thickness on template



Average thickness of 40 people on template

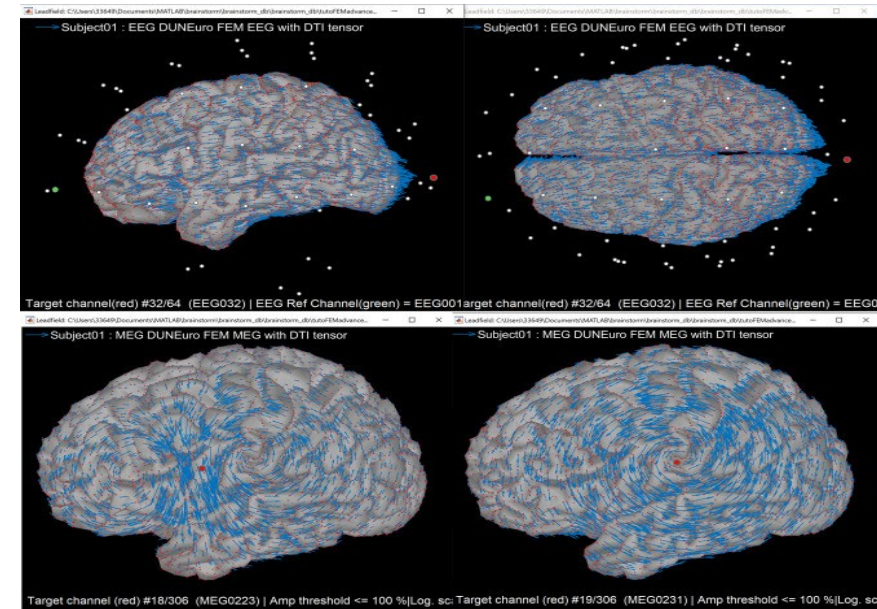


Placing sources for M/EEG localization



Activity from differently aligned and located cortical neuron populations results in different signal in EEG sensors

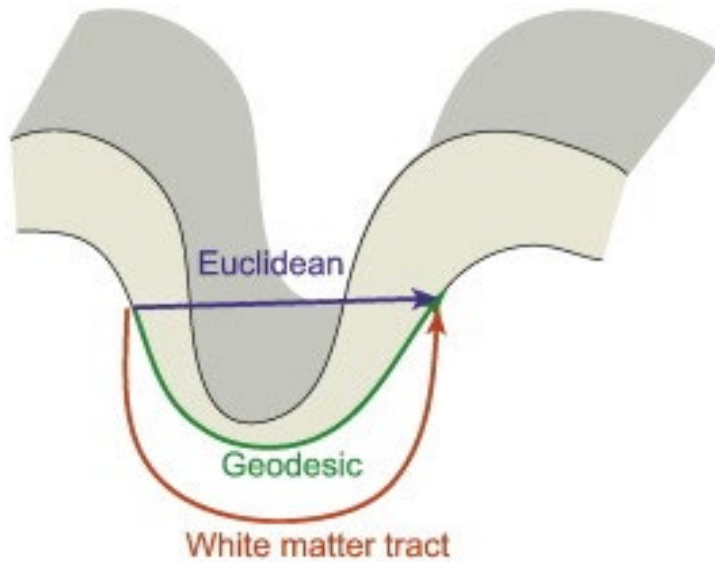
Scherg et al., 2019, *Frontiers in Neurology*, vol. 10



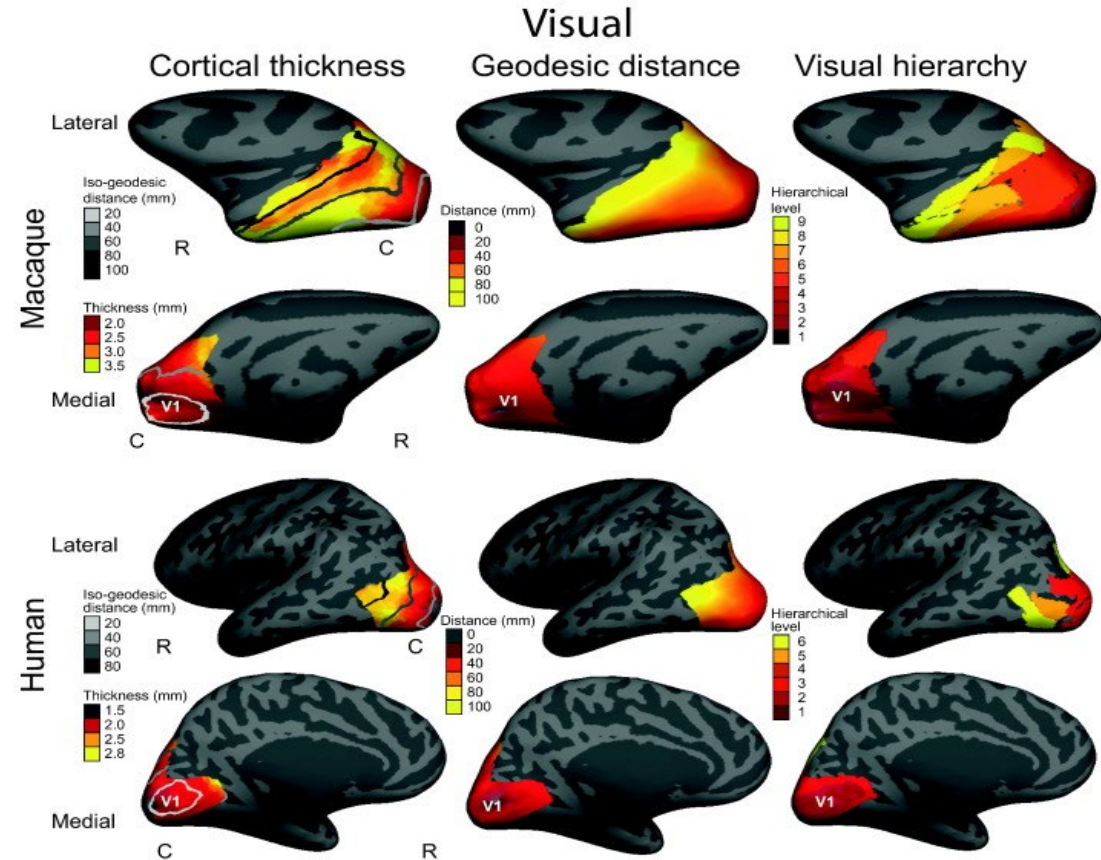
Place current dipoles on the cortical surface and invert to find sources contributing to the sensor measurements

Medani et al., 2023, *NeuroImage*, vol. 267

Geodesic distance measurement



Euclidean distance underestimates the distance between two cortical regions



Geodesic distance (middle column) correlates with cortical organization, i.e., with thickness (first column) and with functional fMRI measurements of the organization of visual cortex (last column).

Wagstyl et al., 2015, *NeuroImage*, vol 111

Visualization for virtual reality augmentation

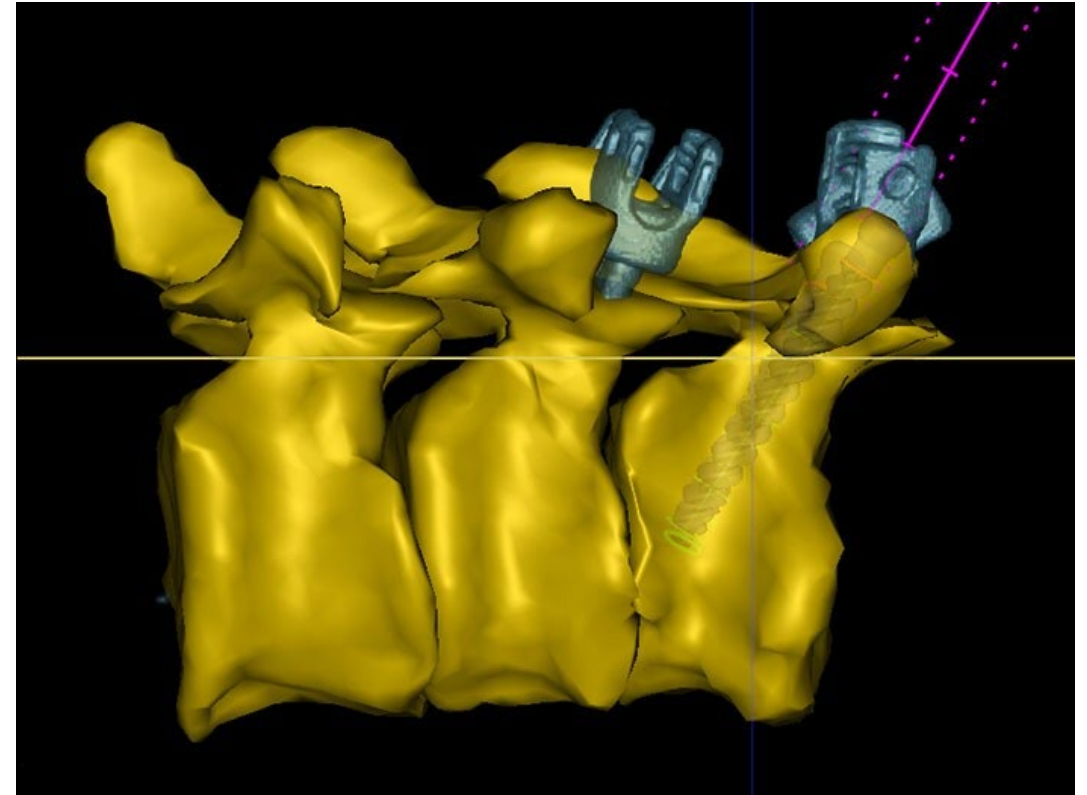
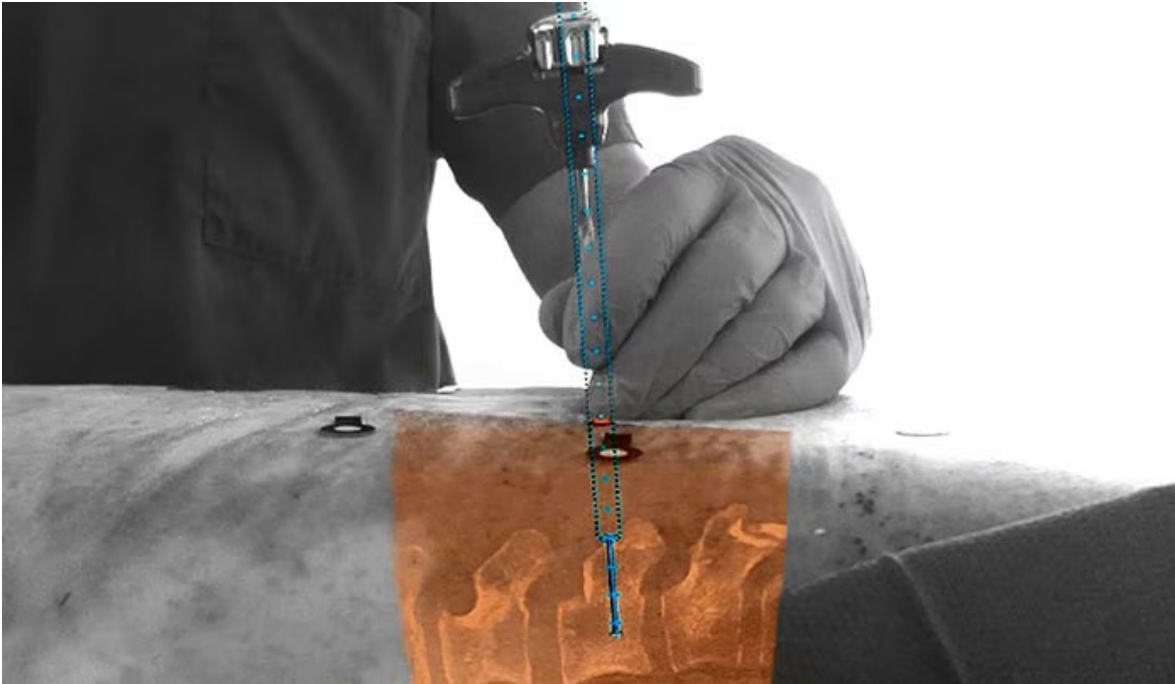
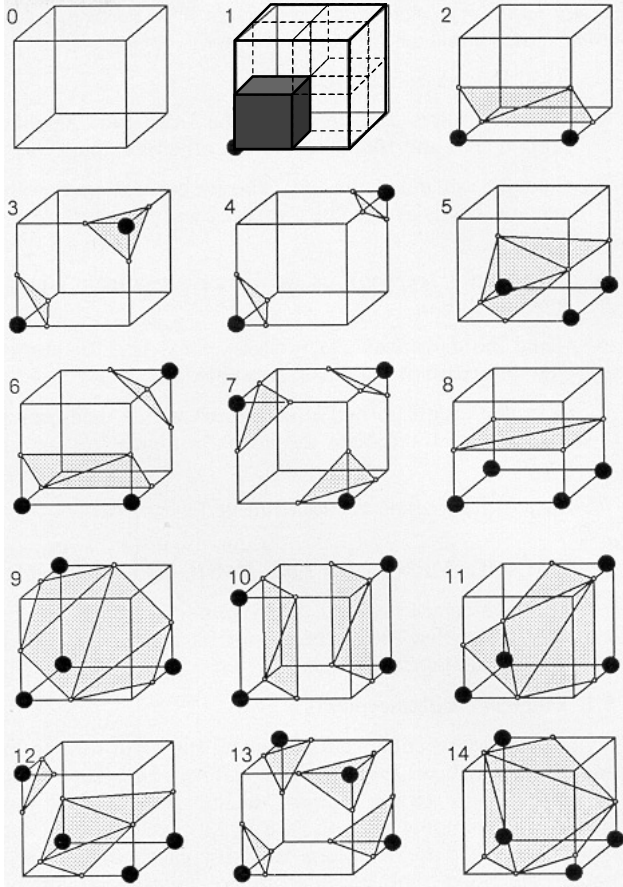


Image-guided spinal surgery navigation with augmented reality

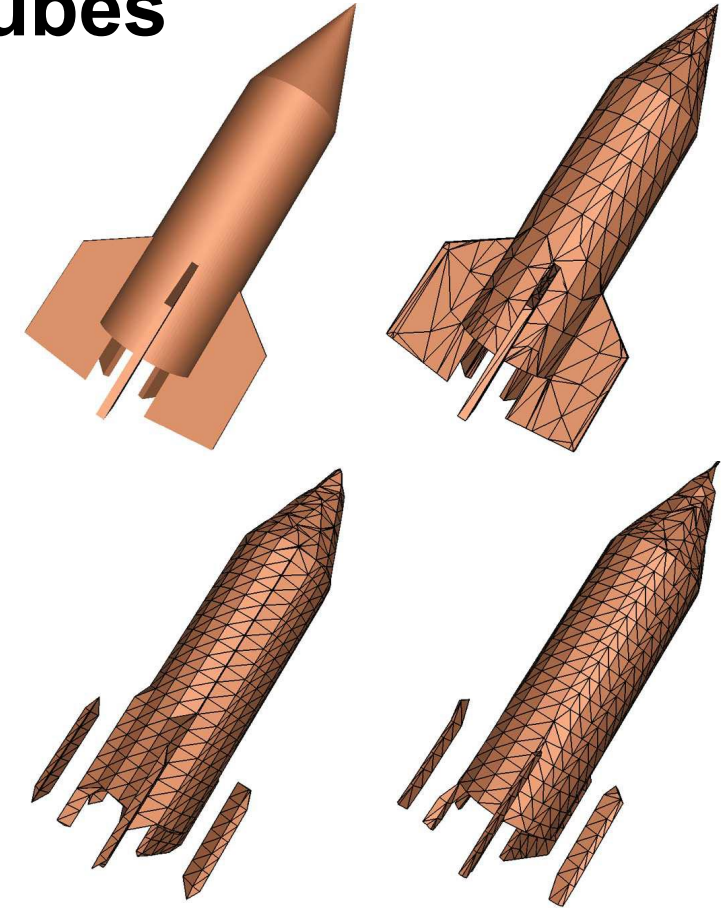
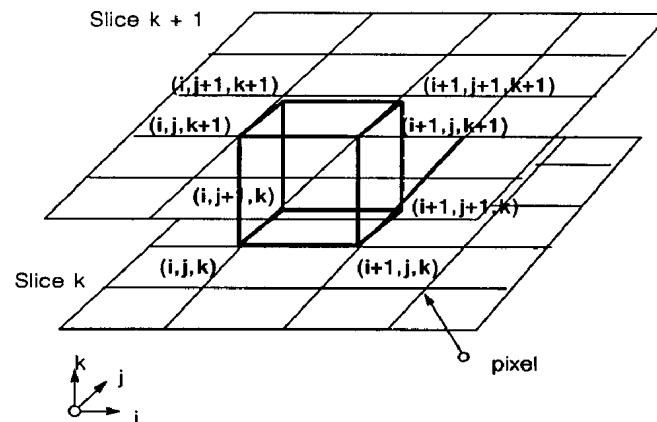
<https://www.philips.com/a-w/about/news/archive/standard/news/press/2017/20170112-philips-announces-new-augmented-reality-surgical-navigation-technology-designed-for-image-guided-spine-cranial-and-trauma-surgery.html>

Surface Reconstruction: Marching Cubes



William E. Lorensen, Harvey E. Cline
ComputerGraphics 21(4), 1987

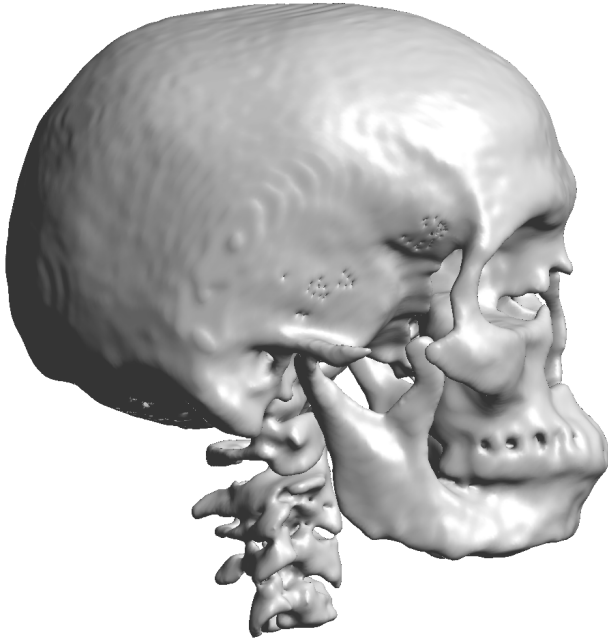
- Start from binary mask
- Voxel centers will be either inside or outside a cube $\rightarrow$ 256 possibilities
- 256 reduce to 14 when using symmetries
- Newer versions reduce number of triangles and reconstruction of thin structures from non-voxel data, or several tissue types



Scott Schaefer and Joe Warren
Dual Marching Cubes: Primal Contouring of Dual Grids
IEEE Trans Vis Comput Graph 2007 13(3):610-9

Smoothing

Taubin

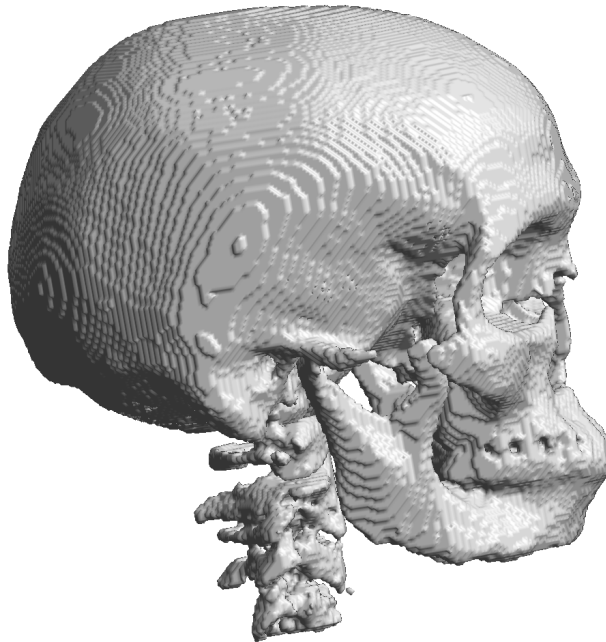


combines 2 Gaussian smoothing steps,
but with opposite signs,
iterated N times

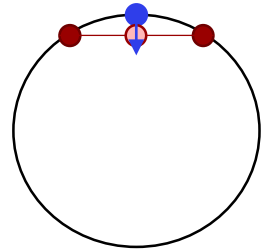
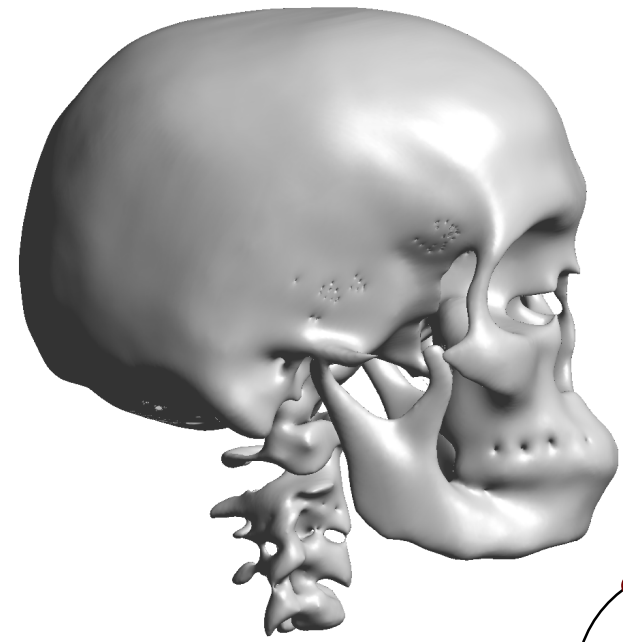
$$X^N = ((I - \mu K)(I - \lambda K))^N X$$

Taubin, 1995

$$0 < \lambda < -\mu$$



Gaussian



$$X^N = (I - \lambda K)^N X$$

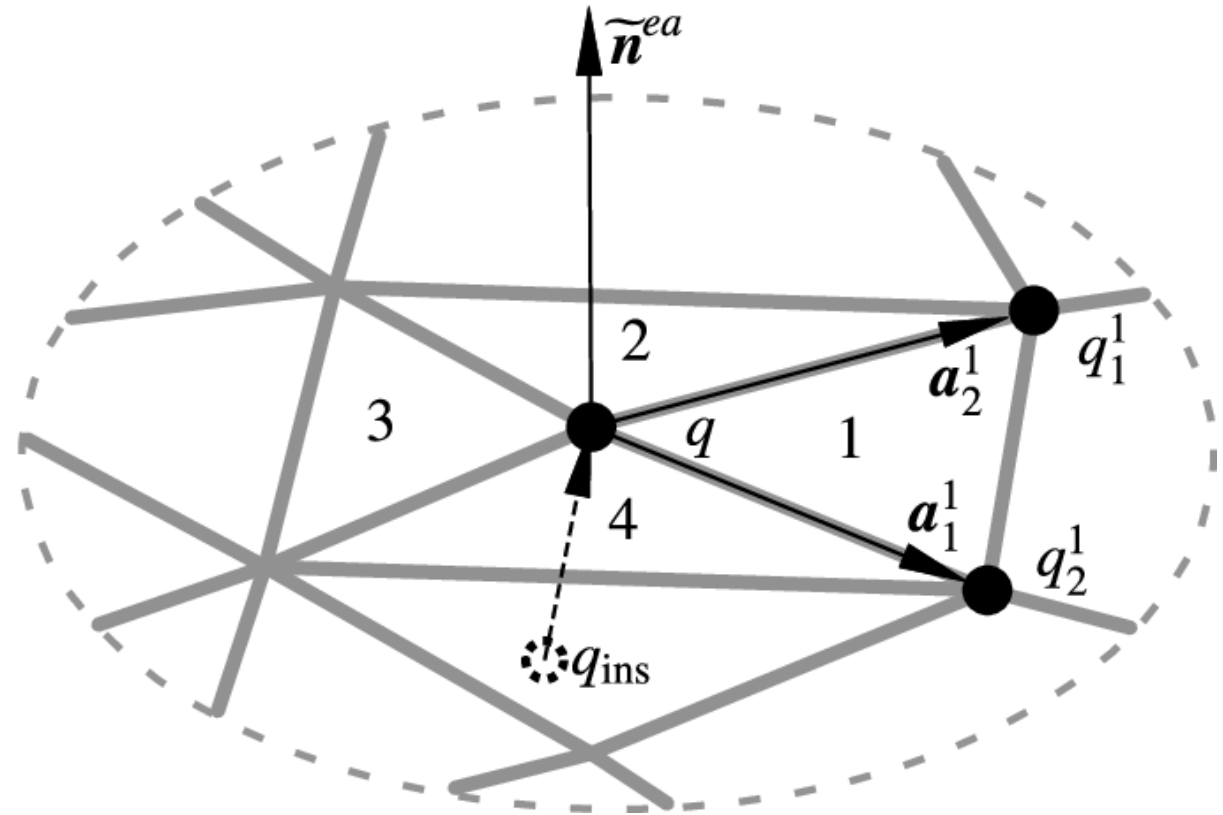
Calculating surface normals



Cross-product to get the normal direction
given vectors **a** and **b**:

$$\mathbf{a} \times \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \sin(\theta) \mathbf{n} \quad \text{or}$$

```
>>> x = [1, 2, 3]
>>> y = [4, 5, 6]
>>> np.cross(x, y)
array([-3,  6, -3])
```



On a surface take the cross-product of edge vectors.

Thank you for your attention